Power BI Project Documentation

Adidas Sales Analysis

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1. Project Objective:

The primary objective of this project is to analyze Adidas sales data using Microsoft Power BI to generate meaningful business insights and support data-driven decision-making. The project transforms raw sales data into interactive dashboards and visualizations for better understanding of business performance. It analyzes key metrics such as total sales, profit, revenue, units sold, and operating margin to identify sales trends and high-performing products. The dashboards allow comparison of sales across regions, retailers, product categories, and time periods. Advanced visualizations, filters, and KPI indicators help stakeholders interpret data effectively and make informed decisions. Additionally, the project identifies seasonal demand patterns, underperforming areas, and profitable segments to support strategic planning and sales optimization.

2. Project Aim:

Monitor overall sales performance and profitability.

Analyze regional and product-wise sales trends.

Identify high-performing and low-performing products and markets.

Evaluate pricing strategies and profit margins.

Support data-driven business decisions through KPI tracking and visual analytics.

Improve inventory planning and sales optimization strategies.

3. Project Overview

The Adidas Sales Analysis project focuses on transforming raw sales data into meaningful business intelligence using Power BI. The dashboard will help stakeholders understand sales behavior, customer demand, retailer performance, and profitability trends across different regions and products.

The project includes:

Data cleaning and transformation

Data modeling

KPI creation

Dashboard development

Business insights generation

4. Problem Statement:

Adidas wants to evaluate its historical sales performance across multiple business dimensions such as:

Products

Retailers

Regions

Sales methods

Profitability

Pricing

Units sold

Currently, decision-makers lack a centralized analytical system to monitor performance effectively. Therefore, an interactive Power BI dashboard is required to:

Track KPIs in real time

Identify sales opportunities

Improve profitability

Enhance strategic planning

5. Business Requirements

The business requires a dashboard that provides:

Clear understanding of sales trends and business performance.

Identification of profitable and non-profitable regions.

Product demand analysis for inventory planning.

Retailer performance evaluation.

Margin and pricing analysis for profitability improvement.

Interactive filtering and drill-through capabilities.

Executive-level KPI summaries and detailed reports.

6. Tools & Technologies Used

Tools	Purpose
Microsoft Powerbi	Data Visualization and Dashboard Creation
Excel/CSV Dataset	Data Source
Power Query	Data Cleaning and Transformation
DAX	Data Calculation and KPI Generation

7. Dataset Description

The dataset contains Adidas sales-related information including:

- Invoice Date
- Region
- State
- City
- Retailer
- Product
- Sales Method
- Price per Unit
- Units Sold
- Total Sales
- Operating Profit
- Operating Margin

8. Data Cleaning & Preparation

The following preprocessing steps were performed in Power Query:

- Removed duplicate records
- Handled missing/null values
- Corrected data types
- Created Month and Day columns from Invoice Date
- Standardized retailer and product names
- Formatted currency columns

9. DAX Measures Used

NAME	DAX MEASURE	PURPOSE
Total Sales	`Total Sales Amt = SUM('Data Sales Adidas'[Total Sales])`	Calculates total revenue generated from sales.
Total Units sold	`Total Units Sold = SUM('Data Sales Adidas'[Units Sold])`	Calculates total quantity of products sold.
Total Profit	`Total profit = SUM('Data Sales Adidas'[Operating Profit])`	Calculates overall operating profit.
Profit Margin %	`Profit Margin% = DIVIDE( SUM('Data Sales Adidas'[Operating Profit]), SUM('Data Sales Adidas'[Total Sales]), 0`	Measures profitability percentage.
Average Price per Unit	`Avg Price/Unit =`	Calculates average selling

	`AVERAGE('Data Sales Adidas'[Price per Unit])`	price per product unit.
Number of Cities	`No.of cities = DISTINCTCOUNT('Data Sales Adidas'[City])`	Counts total cities involved in sales.
Total Products	`Total products = DISTINCTCOUNT('Data Sales Adidas'[Product])`	Counts unique products sold.
Shoppee Platforms	`Shoppee Platforms = DISTINCTCOUNT('Data Sales Adidas'[Sales Method])`	Counts unique retailers.

10. Dashboard Pages Overview

PAGE 1 – SALES OVERVIEW DASHBOARD

Purpose

Provides a high-level overview of Adidas sales performance including revenue, profit, units sold, regional sales, and sales methods.

Chart 1 – Trend Line of Units Sales

Chart Type

Line Chart

Requirement

Analyze monthly sales trend.

Fields Used

Field	Value
X-Axis	Month

Y-Axis	Total Units Sold

Purpose

- Identifies seasonal trends
- Detects sales growth or decline
- Helps forecasting future sales

Chart 2 – Units Sold vs Sales Category

Chart Type

Pie Chart

Requirement

Analyze contribution of different sales methods.

Fields Used

Field	Value
Legend	Sales Method
Values	Total Units Sold

Purpose

- Shows sales distribution
- Compares Online, Outlet, and In-store performance
- Identifies dominant sales channels

Chart 3 – Region vs Sales

Chart Type

Horizontal Bar Chart

Requirement

Compare sales performance by region.

Fields Used

Field	Value
Y-Axis	Region
X-Axis	Total Sales Amt

Purpose

- Identifies highest-performing regions
- Detects low-performing markets
- Supports regional business planning

Retailer Slicer

Requirement

Enable retailer-based filtering.

Field Used

Retailer

Purpose

- Allows interactive dashboard filtering
- Enables retailer-specific analysis

PAGE 2 – PRODUCT ANALYSIS DASHBOARD

Purpose

Analyzes product performance, retailer contribution, and profitability.

Chart 1 – Products and Sales

Chart Type

Clustered Column Chart

Requirement

Compare sales across products.

Fields Used

Field	Value
X-Axis	Product
Y-Axis	Total Sales Amt

Purpose

- Identifies best-selling products
- Helps inventory planning
- Measures product demand

Chart 2 – Product vs Profit

Chart Type

Horizontal Bar Chart

Requirement

Analyze profitability by product.

Fields Used

Field	Value
Y-Axis	Product
X-Axis	Total Profit

Purpose

- Identifies most profitable products
- Supports pricing strategy
- Detects low-margin products

Chart 3 – Units Sold by Seller

Chart Type

Funnel Chart

Requirement

Compare retailer contribution.

Fields Used

Field	Value
Group	Retailer
Values	Total Units Sold

Purpose

- Measures retailer performance
- Identifies top-selling retailers
- Supports retailer partnerships

Chart 4 – Sales vs Units Sold

Chart Type

Scatter Chart

Requirement

Analyze relationship between sales and quantity sold.

Fields Used

Field	Value
X-Axis	Total Sales Amt
Y-Axis	Count of Units Sold
Legend	Product

Purpose

- Identifies product sales patterns
- Detects high-volume products
- Helps pricing and demand analysis

City Slicer

Purpose

Allows city-wise filtering and analysis.

11. Key Business Insights

- West region generated highest sales revenue.
- Men’s Street Footwear achieved maximum sales.
- Online and Outlet channels contributed major sales share.
- Some products generated high sales but lower profit margins.
- Seasonal fluctuations were visible in monthly sales trends.
- West Gear and Foot Locker were top-performing retailers.

12. Advantages of the Dashboard

- Interactive and user-friendly interface
- Real-time business insights
- Improved decision-making
- Easy trend identification
- Better regional and product analysis

13. Future Enhancements

- Integration with live cloud databases
- AI-based sales forecasting
- Customer segmentation analysis
- Predictive analytics using Machine Learning

- Inventory optimization dashboard
- Mobile-responsive dashboard design
- Drill-through detailed reports
- Real-time data refresh automation
- Integration with ERP/CRM systems

14. Challenges Faced

- Handling inconsistent raw data
- Data cleaning and formatting
- Creating optimized DAX measures
- Maintaining dashboard performance
- Designing effective visual layouts

15. Conclusion

The Adidas Sales Analysis Dashboard developed using Microsoft Power BI successfully transforms raw sales data into meaningful visual insights that support strategic business decisions. The dashboard provides comprehensive analysis of sales, profit, regional performance, product trends, and retailer contributions through interactive visualizations and KPI metrics. By leveraging DAX calculations, filters, slicers, and dynamic charts, the project enables efficient performance monitoring and business analysis. Overall, the project demonstrates the power of business intelligence tools in improving operational efficiency, identifying growth opportunities, and supporting data-driven decision-making within an organization.